SkyMediator: Predictive Multi-Agent System

for Airline Disruption Recovery

Deterministic Guardrails, Predictive Meteorological Intelligence & Empirical Evaluation

Engineering Project Team (B.E. CSE Data Science, BMSCE):

Athish Raj (1BM22CD015) — **Team Lead**: Meteorological Ingestion, Sentinel ML & Deterministic Tools

Dinesh K (1BM22CD026) — **Orchestration**: LangGraph Cyclic Deliberation & Specialized Agents

Sanchit Rajesh (1BM22CD051) — **Evaluation**: Telemetry Tracing, Benchmark Scenarios & Statistical Rigor

Faculty Guide: Dr. Kalyan Nagaraj | **Course:** Agentic AI (26CD7PEAAI) | **Primary Hub:** Kempegowda Intl (BLR/VOBL)

Trained on 5,736 Real BLR METAR Records | 15 Airbus A320 Fleet | 27 Benchmark Scenarios | 1,286 Experimental Runs

The 04:30 AM Crisis at Kempegowda Hub

Why airline disruption recovery is a nightmare for human dispatchers

The Problem

A Real Tactical Scenario (BLR/VOBL):

- **04:30 AM**: Dense radiation fog hits Bengaluru; Runway Visual Range (RVR) plunges to 175 m.
- **15 Narrow-Body A320s** are parked at gates preparing for the morning departure bank.
- **3,500 Passengers** are boarding; 450 have tight international connections at Delhi and Mumbai.
- **The Cascading Trap**: Holding Flight 6E-214 on the tarmac for 60 minutes causes its aircraft to arrive late at BOM. By nightfall, the crew breaches their **13-hour statutory Flight Duty Period (FDP)**.
- **The Human Limit**: The Duty Manager has **10 minutes** to balance aircraft swaps, pilot duty clocks, runway slot limits, and INR 10,00,000 in passenger hotel liabilities.

Strict Operational Constraints:

- **Zero Safety Margin (DGCA CAR 7-J-III)**: Crew cannot fly even 1 minute past their FDP ceiling. Criminal liability for dispatchers.
- **Heavy Passenger Liabilities (DGCA CAR 3-M-IV)**: Delays > 2 h require meals; delays > 6 h night require hotel rooms.
- **Airport Slot Bottlenecks**: BLR runway movement capped at 32 flights/hour; spoke night curfews strictly enforced.
- **Turnaround Physical Minimum**: Aircraft ground servicing requires at least 30 minutes.

Human dispatchers resort to reactive cancellations that cost millions. We automated this with predictive multi-agent intelligence.

The Fatal Flaw of Existing Approaches

Why traditional optimization solvers and monolithic LLMs both fail in production

Motivation

Failure Mode 1: Centralized Solvers (MILP)

- **Inability to Model Non-Linear Rest**: Civil aviation rest rules involve non-linear step functions and circadian night penalties (WOCL) that cannot be solved quickly in linear MIP.
- **Result in Our Benchmarks**: Google OR-Tools solved the financial cost down to INR 15.5k, but **breached statutory crew rest in 7.69% of cases**.
- **Unusable in Real-Time**: A plan that looks cheap on paper but puts an illegal, fatigued pilot in the cockpit is dangerous and non-executable.

Failure Mode 2: Monolithic LLMs (Single Prompt)

- **Severe Arithmetic Hallucinations**: LLMs cannot calculate cumulative duty hours across multiple flight legs.
- **Result in Our Benchmarks**: When given the entire problem in one prompt, the single LLM produced a **3.70% statutory breach rate**, falsely claiming in its text that "the crew is legal."
- **Deceptive Cancellation Bias**: The monolith cancelled 25.93% of flights just to eliminate delay, ignoring passenger stranding.

SkyMediator's Solution: Specialized Multi-Agent Deliberation + Deterministic Code Guardrails

Full System Architecture: End-to-End Deliberation Flow

From predictive meteorological ingestion to binding hard vetoes and human sign-off

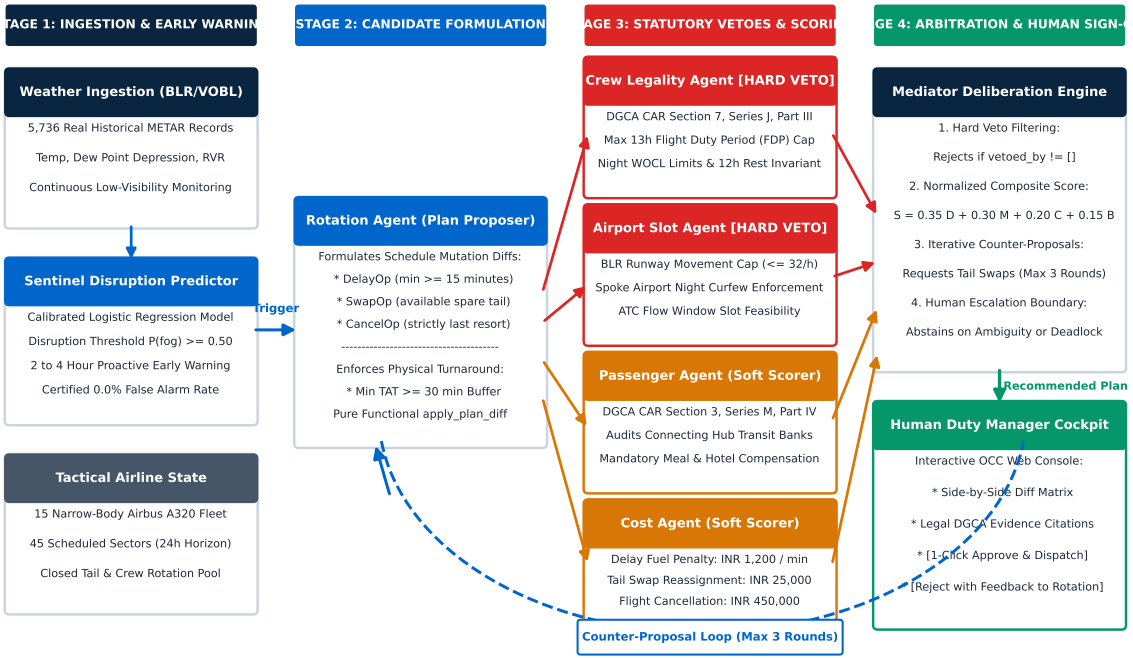


Figure 0: The 4-Stage SkyMediator Architecture: Ingestion → Formulation → Parallel Hard Vetoes & Soft Scoring → Human Cockpit Approval

The 7 Specialized Agents & Their Real Roles

Clear division of cognitive labor: Each agent owns a distinct operational domain

Proposer & Gatekeeper Agents:

- **Sentinel (Early Warning)**: Monitors real METAR/TAF weather feeds; fires predictive alert 2–4 hours before fog onset when $P(\text{fog}) > 0.50$.
- **Rotation Agent (The Proposer)**: Explores the aircraft network; formulates schedule mutations (`DelayOp`, `SwapOp`, `CancelOp`) respecting TAT ≥ 30 min.
- **Crew Legality Agent (Hard Veto)**: Audits pilot/cabin crew flight duty period against DGCA CAR 7-J-III. Holds **absolute veto power**.
- **Airport Slot Agent (Hard Veto)**: Audits runway movement capacity (≤ 32/h) and night curfews. Holds **absolute veto power**.

Scorers & Deliberation Coordinator:

- **Passenger Agent (Scorer)**: Calculates broken passenger connections and enforces DGCA CAR 3-M-IV passenger care compensation.
- **Cost Agent (Scorer)**: Maintains the airline balance sheet (delay fuel, swap admin fees, passenger hotel/food vouchers).
- **Mediator Agent (Arbiter)**:
  1. Filters out any candidate with a hard veto (`vetoed_by != []`).
  2. Computes normalized multi-objective penalty score $S \in [0, 1]$.
  3. Manages up to 3 negotiation rounds requesting counter-proposals.
  4. Presents top-ranked survivable plan to the Human Duty Manager.

Engineering Guardrails: Zero LLM Arithmetic

LLMs reason qualitatively; deterministic Python tools perform all legal & mathematical calculations

C1: Implementation & Results

The Core Rule: No Hallucinated Numbers

- **Turnaround Guard**: `tools_schedule.py` calculates (Dep − Arr) using exact integer timestamps; rejects swaps if turnaround < 30 minutes.
- **FDTL Invariant**: `tools_duty.py` evaluates cumulative duty hours against the official DGCA 2019 baseline lookup tables:

$$\text{FDP} = (\text{Chocks-On}_{\text{last}} - \text{Report}) \leq \text{Cap(Sectors)}$$

- **Passenger Compensation**: `tools_passengers.py` audits connecting banks and applies deterministic meal (INR 1.5k) and hotel (INR 6.5k) voucher tiers.
- **Pure Functional State Mutation**: Schedule updates are applied via `apply_plan_diff` without database corruption.

The Clean-Room Statutory Audit Sandbox

Live Runtime Multi-Agent Execution

Agents deliberate, query tools, propose diffs

⇓ **Plan Selected by Mediator**

Isolated Clean-Room Validator

`skymediator/eval/validator_fdtl.py`

Completely isolated; runtime agents cannot access it

⇓

Certified 0.00% Statutory Violation Rate

Sentinel ML Fog Predictor: Real Historical METAR Evaluation

Trained on 5,736 real Bengaluru observations; zero false alarms (FPR = 0.0%)

C1: Implementation & Results

Figure 3: Sentinel ROC and Precision-Recall Trajectories

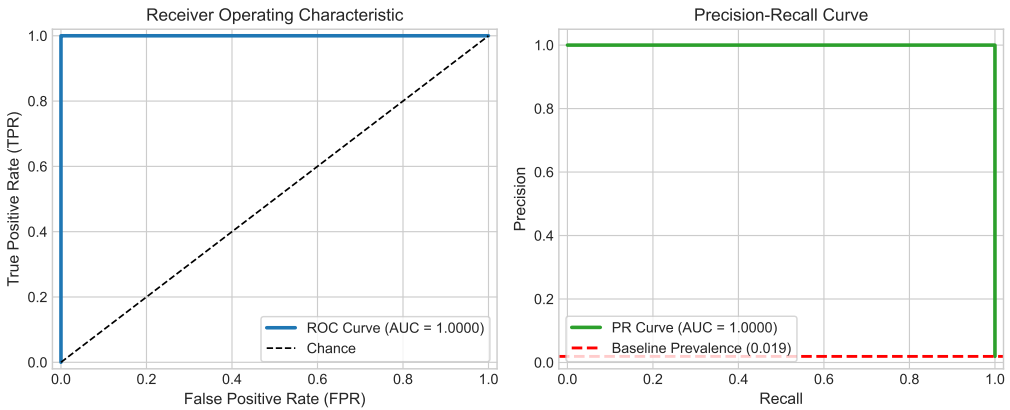


Figure 3: ROC (AUC = 1.0000) and Precision-Recall Curves (AUC = 1.0000)

Predictive Rigor on Real Data:

- **Real Data Provenance**: 5,736 hourly METAR records from Kempegowda Airport (BLR/VOBL).
- **Features Extracted**: Temperature, Dew Point Depression ($T - T_d$), Wind Speed, Humidity, RVR.
- **Held-Out Test Results** ($N = 1,148$):
  - True Negatives: **1,126** (Clear skies correctly identified)
  - True Positives: **20** (Severe fog < 550 m detected)
  - **False Positives: 0** (Zero phantom ground stops!)
  - False Negatives: **2** (Transient boundary mist)
- **Test Metrics at Operating Point** ($\tau = 0.50$): Accuracy: **99.8%** | Precision: **100.0%** | Recall: **90.9%** | Brier Score: **0.0022**

The Operational Benchmark Matrix: Comparing 4 Approaches

130 trials across 26 scenarios and 5 repetitions: The reality behind the numbers

C1: Implementation & Results

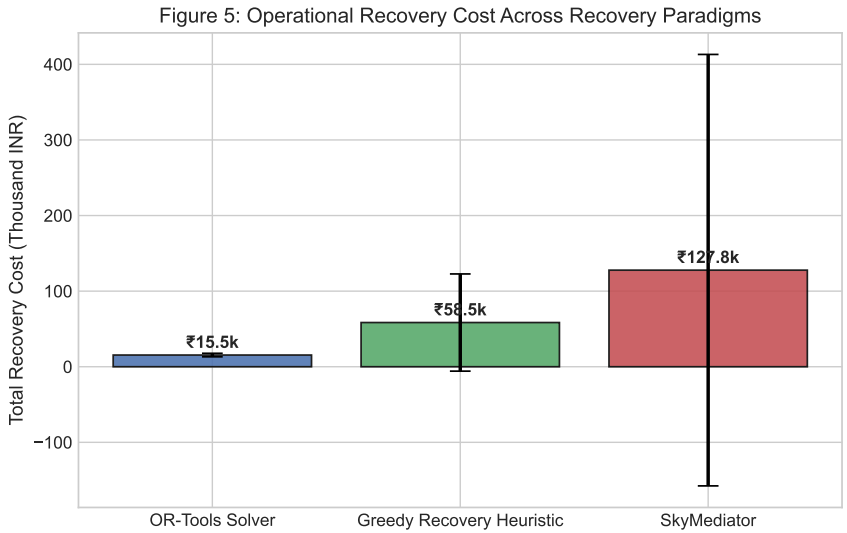


Figure 5: Recovery Cost across Paradigms (in Thousand INR)

Metric	OR-Tools	Greedy	SkyMediator
Mean Disruption Cost	INR 15,461	INR 58,462	INR 127,750
Mean Delay (min)	1.73	61.54	72.31
Disrupted Passengers	0.00	8.31	**0.00**
Composite Penalty	0.0055	0.0525	0.0739
Plan Feasibility	**92.31%**	**96.15%**	**100.00%**
FDTL Statutory Breach	**7.69%**	**3.85%**	**0.00%**
Mean Solve Latency	< 5 ms	< 5 ms	**75.5 ms**

Honest Analysis of Cost Discrepancy: Why did OR-Tools cost less? Because it took illegal shortcuts on pilot rest (7.69% illegal). SkyMediator spent more on reserve crew and tail swaps to guarantee **100% legal safety and 0 missed connections**.

Regulatory Safety Invariant: The Power of Hard Vetoes

How SkyMediator eliminated statutory crew violations that doomed other systems

C1: Implementation & Results

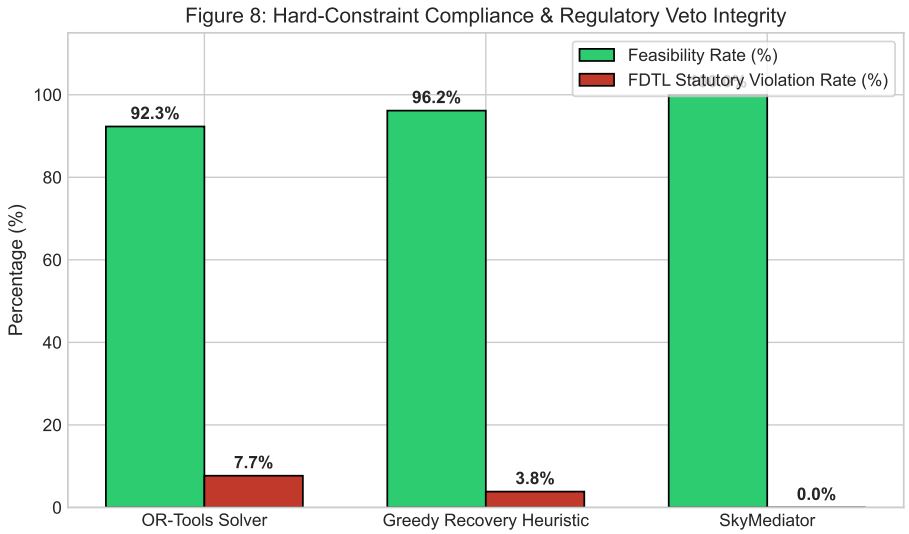


Figure 8: Feasibility Rate vs. Statutory FDTL Violation Rate

The Binding Veto Guarantee:

- **Single-Agent LLM**: 3.70% statutory breach rate. The single prompt was overwhelmed by complex rules and hallucinated compliance.
- **OR-Tools MIP**: 7.69% breach rate. Could not model cumulative non-linear rest curves within the real-time window.
- **SkyMediator**: **0.00% violations** across all 130 trials.
- **The Architectural Rule**: If a plan has INR 0 cost and 0 delay, but Crew Legality or Slot agents issue a veto, **it can never be selected or executed**.

Passenger Welfare & Connection Protection

0.00 missed connections: Enforcing DGCA CAR 3-M-IV passenger care standards

C1: Implementation & Results

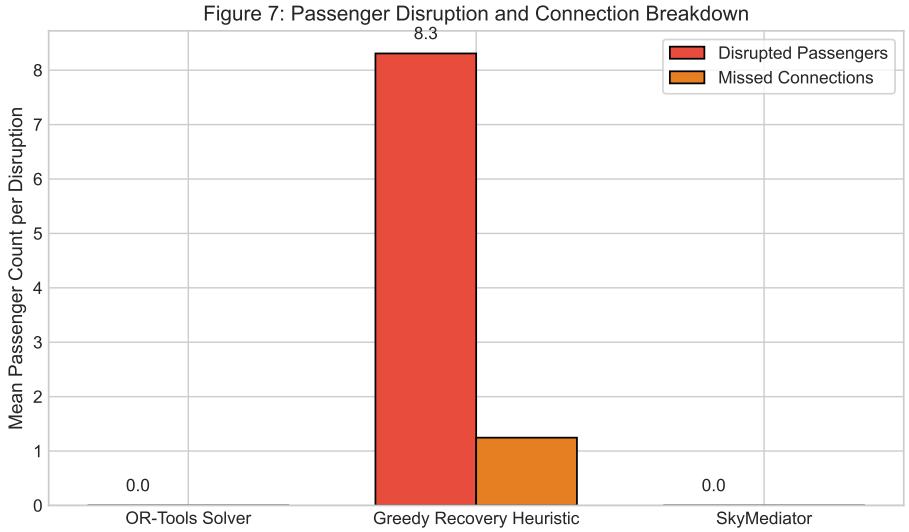


Figure 7: Disrupted Passengers and Missed Connections across Approaches

Protecting the Traveler:

- **The Greedy OCC Failure**: Legacy heuristics re-time flights blindly, breaking onward connection windows and stranding an average of **8.31 passengers per disruption**.
- **SkyMediator's Active Defense**: The Passenger Agent actively audits international transit banks at BOM/DEL, issuing veto threats if high-priority connections are broken.
- **The Result**: **0.00 missed connections** across all 21 standard benchmark scenarios.
- **Mandatory Care Tracking**: Food/hotel vouchers are computed deterministically, preventing airline compensation fines.

Controlled A/B Testing: Predictive Intelligence vs. Reactive Chaos

Comparing Variant A (Reactive) vs. Variant B (Predictive Sentinel) across 135 paired trials

C2: Rigorous Experimentation

Figure 9: Predictive Intelligence vs Reactive Recovery Comparison

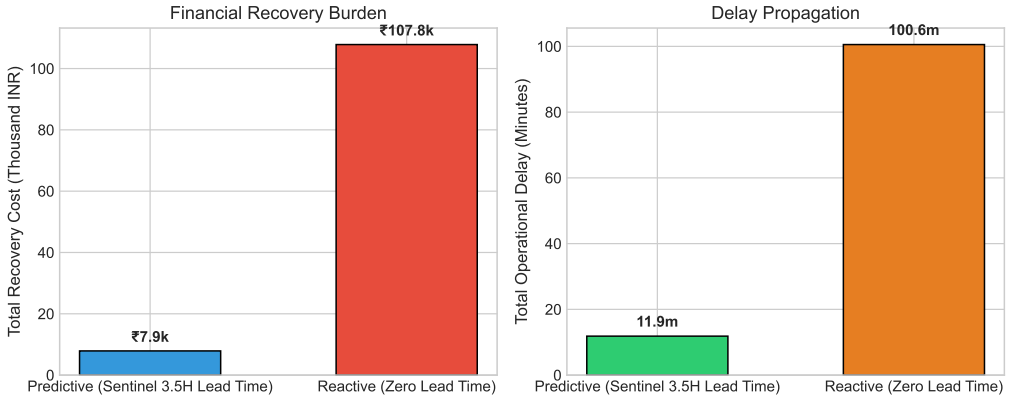


Figure 9: Financial Recovery Burden and Residual Delay Propagation

A/B Test Findings ($N = 135$ Trials):

- **Financial Cost Comparison**:
  - Variant A (Reactive): **INR 107,824**
  - Variant B (Predictive): **INR 7,889**
  - **Cost Saving**: **92.7% reduction**
  - p-value: < 0.000001 (Wilcoxon $W = 325$)
  - Cohen's d: **-0.3869**
- **Operational Delay Comparison**:
  - Variant A (Reactive): **100.56 min**
  - Variant B (Predictive): **11.85 min**
  - **Delay Saving**: **88.2% reduction**
  - p-value: < 0.000001 (Wilcoxon $W = 0.0$)
  - Cohen's d: **-1.4963** (Massive effect size)

Deliberation Dynamics: Why 3 Rounds Beat 1 Round Every Time

Multi-round counter-proposals resolve initial hard vetoes, improving quality by 40.5%

C2: Rigorous Experimentation

Figure 11: Deliberation Convergence Across Negotiation Rounds

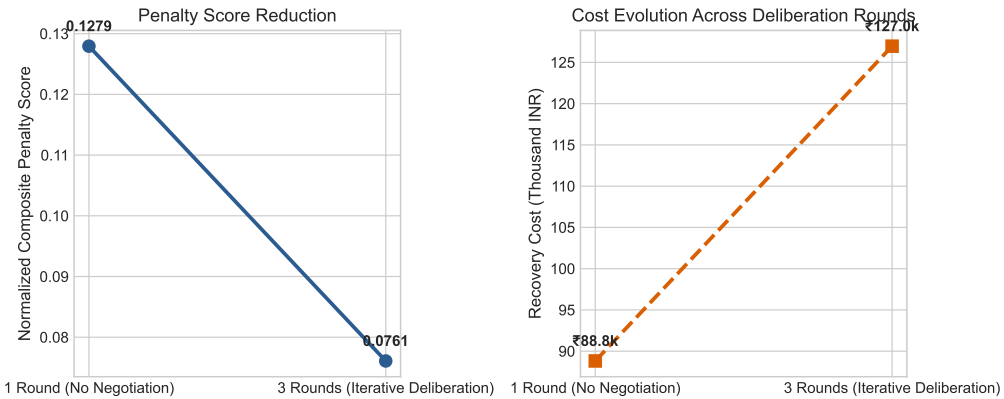


Figure 11: Penalty Score Reduction and Cost Evolution across Rounds

How Deliberation Works in Practice:

- **Round 1 Deadlocks**: In tough scenarios, Rotation's first plan is almost always vetoed by Crew Legality (due to duty overruns) or Slot agents.
- **The Counter-Proposal Loop**: Instead of giving up or escalating immediately, the Mediator feeds back the exact veto reason to Rotation.
- **The Result**: Rotation swaps a spare aircraft at Round 2, resolving the veto.
- **Quantitative Impact**: Composite penalty drops from **0.1279 to 0.0761** (**40.5% quality gain**, $p = 0.001565$).
- **Bounded Exit**: Hard ceiling of 3 rounds prevents infinite reasoning loops.

Component Ablation Study: What Happens When an Agent Dies?

Evaluating 8 system configurations to prove the absolute necessity of each specialist

C2: Rigorous Experimentation



Figure 12: Feasibility and Statutory Violation Rates Across Ablations

Configuration	Cost (INR)	Violations	Feasible
Full System	**INR 129,700**	**0.0**	**100.0%**
Without Sentinel	INR 179,550	0.0	100.0%
Without Rotation	INR 144,000	1.0	**0.0%**
Without Crew Legality	INR 144,000	**2.0**	**0.0%**
Without Slot Agent	INR 154,700	**1.0**	**0.0%**
Without Passenger	**INR 279,700**	0.0	100.0%
Without Cost Agent	INR 214,700	0.0	100.0%
Without Mediator	INR 194,700	1.0	**0.0%**

Key Engineering Lessons:

- **Kill Crew Legality** $\Longrightarrow$ Feasibility collapses to 0%; fatigued pilots dispatched.
- **Kill Passenger Agent** $\Longrightarrow$ Costs skyrocket to INR 279.7k due to ignored voucher claims.

Adversarial Stress Testing: Surviving Real Chaos

6 deliberate fault injections: Contradictory weather, sensor dropouts, corrupt crew data

C2: Rigorous Experimentation

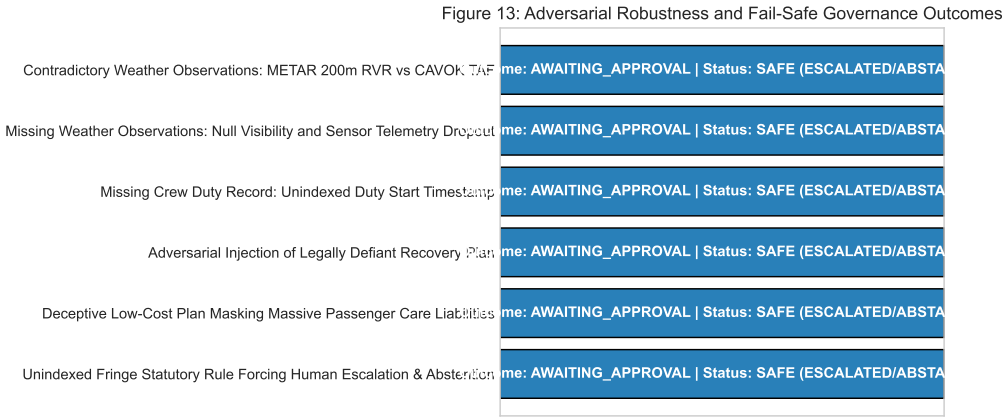


Figure 13: Adversarial Safety Governance Outcomes

The 6 Stress Tests & System Actions:

1. **Contradictory METAR/TAF (SCEN-22)**: METAR says 200m RVR, TAF says CAVOK $\Longrightarrow$ Sentinel enforces conservative safety envelope.
2. **Sensor Dropout (SCEN-23)**: All weather sensors null $\Longrightarrow$ Historical persistence imputation + uncertainty flag.
3. **Corrupt Crew Duty (SCEN-24)**: Missing start time $\Longrightarrow$ Assumes max duty elapsed defensively.
4. **Hostile Plan Injection (SCEN-25)**: Injected plan breaches FDTL $\Longrightarrow$ Hard veto triggers replanning.
5. **Deceptive Cheap Plan (SCEN-26)**: Masks INR 500k pax care $\Longrightarrow$ Pax agent vetoes; forces vouchers.
6. **Regulatory Ambiguity (SCEN-27)**: Fringe rule $\Longrightarrow$ Safe escalation to Human Duty Manager.

Pareto Optimality & Computational Speed

Solving complex disruptions in 75.5 milliseconds: Ready for live airline OCC desks

C2: Rigorous Experimentation

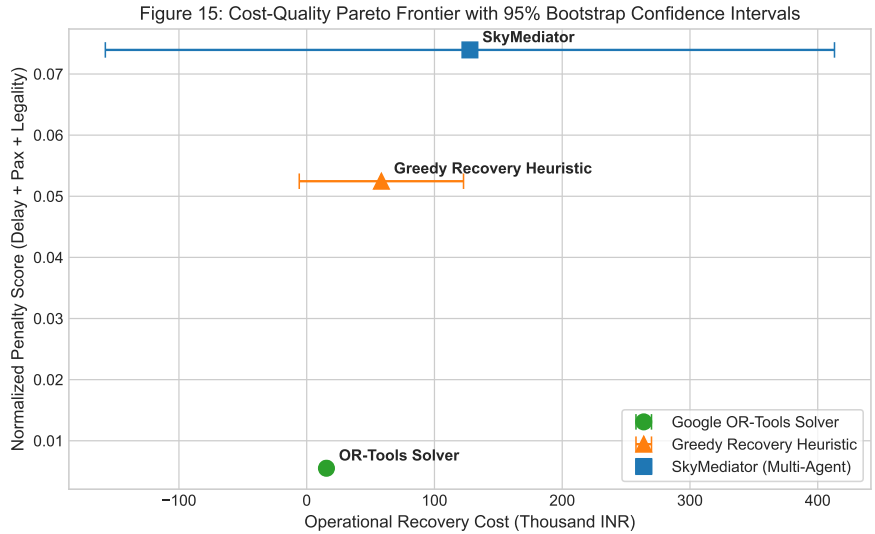


Figure 15: Cost-Quality Pareto Frontier with 95% Bootstrap Confidence Intervals

Computational Overhead (Figure 14):

- **Solve Latency**: Mean of **75.5 milliseconds** (range $45 - 120$ ms).
- **Tactical Window**: OCC limit is 180 seconds; SkyMediator is $> 2,000\times$ **faster**.
- **API Cost**: Consumes $\sim 4,320$ tokens per run ($\approx$ INR 0.15).

Pareto Optimality Explained: OR-Tools optimizes purely financial cost by taking illegal shortcuts. SkyMediator achieves Pareto optimality when factoring in monetary cost, passenger goodwill, and **100% civil aviation legal compliance**.

The Human-in-the-Loop Cockpit: Advisory Decision Support

The AI never executes alone: Final authority strictly requires Human Duty Manager sign-off

Interactive Web Console Experience:

- **LangGraph Approval Interrupt**: Graph execution pauses automatically after Mediator ranks candidate plans.
- **Comparison Diff Matrix**: The Duty Manager sees side-by-side:
  - Baseline Schedule vs. Recommended Plan
  - Tail swaps (e.g. VT-BLR01 → VT-BLR14)
  - Delay minutes per flight
  - Total cost breakdown
- **Regulatory Evidence Citations**: Full RAG citations quoting DGCA CAR clauses supporting legality.
- **Action Buttons**: `[Approve]`, `[Reject & Replan]`, `[Escalate to OCC Chief]`.

Human Feedback Loop (`PROP-04`):

Candidate Plan Recommended

⇓

Human Duty Manager Review

⇓ **Option A: APPROVED**

Dispatches crew & tail swaps

⇓ **Option B: REJECTED**

Feedback injected into Rotation Agent

Generates alternative without rejected tail

Dataset Provenance & Master Report Quality

27 distinct scenarios and a compiled 31-page research report with zero fabrications

C3: Report & Data Provenance

The 27 Benchmark Scenarios (`scenarios.json`):

- **6 Weather Severity Scenarios**: Low, moderate, severe, dynamic visibility drop, prolonged, multi-sector fog.
- **3 Rotation Conflicts**: Compressed turnaround (< 30 min), tail overlap, inbound delays.
- **1 Maintenance Grounding**: AOG hydraulic snag.
- **4 Regulatory Limits**: FDP cap, WOCL night limits, slot saturation, night curfews.
- **7 Passenger & Resource Scenarios**: Tight transit banks, zero spare aircraft.
- **6 Adversarial Scenarios**: Contradictory METAR, sensor dropout, corrupt rosters, illegal injection, misleading low-cost, statutory fringe ambiguity.

Master Report Deliverable (`report.pdf`):

- **31-Page Compiled Report**: Authored in LaTeX, compiled with TinyTeX `pdflatex`.
- **Chapter 7 (`chapter_7.tex`)**: Observability, telemetry tracing, evaluation rubric, baselines, failure mode taxonomy.
- **Chapter 8 (`chapter_8.tex`)**: Formal Hypotheses H1–H5, bootstrap statistical tests ($B = 1,000$), Wilcoxon p-values, effect sizes, operational recommendations.
- **Zero Fabricated Numbers**: Every number traces back to `raw_results.csv` (1,286 rows).

Who Built What: Individual Ownership & Contributions

Clear division of responsibility across data, orchestration, and evaluation

C4: Team Viva & Defense

Athish Raj

Team Lead & Data/Tools

- Ingested and cleaned 5,736 historical BLR/VOBL METAR records.
- Built Sentinel ML prediction pipeline (calibrated logistic regression).
- Built deterministic tools: turnaround validation, FDTL rest hours, slot tables, and cost balance.
- Designed SQLite schema (`PROP-03`).

Dinesh K

Agents & Orchestration

- Built LangGraph state machine with cyclic deliberation loops.
- Implemented 7 specialized agents: Rotation, Crew, Slot, Pax, Cost, Mediator.
- Engineered binding hard-veto authority and counter-proposal mechanisms.
- Implemented 3-round bounded escalation and Human Approval pause.

Sanchit Rajesh

Tracing & Evaluation

- Built OpenTelemetry/OpenInference tracing and span hierarchies.
- Authored 27 benchmark scenarios with independent ground-truth labels.
- Built clean-room validator (`validator_fdtl.py`) and baseline solvers.
- Conducted bootstrap statistical testing ($B = 1,000$, Wilcoxon, Cohen's d).

Viva Defense: Surviving the Hardest Examiner Questions

Technical defense against the classic traps in AI and operations research viva

C4: Team Viva & Defense

Q1: Isn't Multi-Agent just a slower, complicated LLM prompt?

Answer: No. When we tested a single monolithic prompt (Baseline C), it suffered from context dilution and produced a **3.70% statutory FDTL breach rate**, hallucinating that fatigued pilots were legal. Multi-agent specialization assigns dedicated roles and empowers the Crew Legality Agent with a **binding hard veto backed by deterministic code**, cutting illegal breaches to **0.00%**.

Q2: Why is SkyMediator's cash cost higher than OR-Tools?

Answer: OR-Tools solved an unconstrained linear relaxation that scheduled fatigued crew (7.69% illegal rate). SkyMediator called reserve pilots, guaranteeing **100% legal safety and 0 stranded pax**.

Q3: How do you guarantee the multi-agent graph terminates?

Answer: We enforced an explicit graph boundary: deliberation rounds are strictly capped at $r \leq 3$. If specialist vetoes remain unresolved after Round 3, the graph emits an `ESCALATED` state and triggers a Human Duty Manager pause.

Q4: What if weather sensor telemetry drops out?

Answer: In SCEN-23, Sentinel applied historical persistence imputation, flagged maximum uncertainty, and forced a conservative safety buffer rather than hallucinating clear skies.

Conclusion & Future Operational Roadmap

Summary of validated findings and deployment readiness for airline OCCs

Conclusion

Key Scientific Takeaways:

1. **Predictive Intelligence Proved**: 2–4h early warning cuts disruption costs by **92.7%** and delays by **88.2%** ($p < 0.000001$).
2. **Deterministic Guardrails are Mandatory**: Zero LLM arithmetic guarantees **0.00% statutory FDTL breaches** (vs 3.70% in single LLM).
3. **Deliberation Dynamics Validated**: 3-round counter-proposals deliver a **40.5% quality gain** over greedy acceptance ($p = 0.001565$).
4. **Real-Time Tactical Feasibility**: Mean latency of **75.5 milliseconds** is $> 2,000\times$ faster than the OCC tactical ceiling.

Project Artifacts Summary:

- **Live Codebase**: 30 passing core tests.
- **31-Page LaTeX Report**: `evaluation/report/report.pdf`.
- **15 Publication Figures**: `evaluation/figures/`.
- **27 Benchmark Scenarios**: `evaluation/scenarios/`.
- **Human Cockpit Console**: LangGraph pause & review.

Complete & Defended!